JAVA QUIZ APPLICATION

Modules and Features

BCA Project Documentation

Project Overview

The Java Quiz Application is a desktop-based application using Java Swing and an Oracle database. It allows users to log in, select quiz categories, answer multiple-choice questions, calculate scores, save results, and review quiz history. Java Database Connectivity (JDBC) connects the application to Oracle.

1. Login Module

Purpose: Authenticate existing users and register new users before they access the application.

- Enter user/student name.
- Enter login password.
- Enter college name.
- Validate credentials for an existing user.
- Register a new user and save user details in the database.
- Update college name for an existing user where applicable.
- Clear login form fields.

2. Main Menu Module

Purpose: Provide navigation to the application features.

- Open the quiz module.
- Open category management.
- Open question management.
- Open reports/history.
- Open result management.
- Exit or log out, where implemented.

3. Category Management Module

Purpose: Organize quiz questions into categories.

- Add a category.
- View available categories.
- Edit a category, if implemented.
- Delete a category, if implemented.
- Associate questions with the correct category using category IDs.

Categories discussed for this project: Easy, Hard, Medium, and DBMS.

4. Question Management Module

Purpose: Maintain multiple-choice questions for each category.

- Add a question.
- Enter four answer options (A, B, C, and D).
- Set the correct answer.
- Associate the question with a category.
- View stored questions.
- Edit or delete questions, if implemented.
- Retrieve questions from Oracle for a quiz.

5. Quiz Module

Purpose: Let a logged-in user attempt a quiz in a selected category.

- Select a quiz category.
- Load questions and options from the database.
- Select an answer from the available options.
- Check selected answers against the correct answers.
- Calculate correct and wrong answer counts.
- Calculate score percentage.
- Save the result with user and category information.
- Return to the main menu after the quiz, where implemented.

6. Result Management Module

Purpose: Store and display quiz outcomes.

- Store result ID, user ID, category ID, total questions, correct answers, wrong answers, score percentage, and result date.
- View a user's own results.
- View all users' results in the broader history view, if enabled.
- Display username, college name, category, score, and date where supported by the report screen.
- Delete a result, if implemented.

7. Report / History Module

Purpose: Show past quiz attempts and make results easier to review.

- View the logged-in user's quiz history.
- View all users' quiz history in the all-history view, if enabled.
- Order history by the most recent result date.
- Display category, total questions, correct answers, wrong answers, score percentage, and result date.
- Include username and college name in the all-users history view where available.
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